Heart Attack Risk Prediction Using Ordered Logistic Regression Using R

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KNOWLEDGE - COMMITMENT - LEADERSHIP

Introductio

n **Ordinal Logistic Regression** is a statistical method used to predict the relationship between one or more independent variables (predictors) and a dependent variable that has ordered categories.

1 **Ordered Logistic Regression**

Used to predict ordinal variables.

2 **Project Objective**

Predict heart attack risk levels.

Dataset Overview

Variable	Description
Age	Age of the individual.
Sex	0 (Female), 1 (Male).
Chest Pain Type	0: Typical Angina, 1: Atypical Angina, 2: Non-anginal Pain, 3: Asymptomatic.
Resting Blood Pressure	Blood pressure at rest (mm Hg).
Cholesterol	Serum cholesterol level (mg/dL).
Fasting Blood Sugar	0 (<=120 mg/dL), 1 (>120 mg/dL).
Resting ECG	0: Normal, 1: Abnormal, 2: Hypertrophy.
Max Heart Rate	Maximum heart rate achieved during exercise.
Exercise-Induced Angina	0 (No), 1 (Yes).
Oldpeak (ST Depression)	Depression in ECG during exercise (mm).
Slope of ST Segment	0: Upsloping, 1: Flat, 2: Downsloping.
Num Vessels Fluoroscopy	Number of major vessels (0 to 3).
Thalassemia	0: Normal, 1: Fixed Defect, 2: Reversible Defect.
Risk Level	0 (Low), 1 (Medium), 2 (High) – Target variable for heart attack risk.

Exploratory Data Analysis (EDA)

Understanding data exploring our data

Number of Rows	100
Number of Columns	14
Dimension of dataset	100, 14

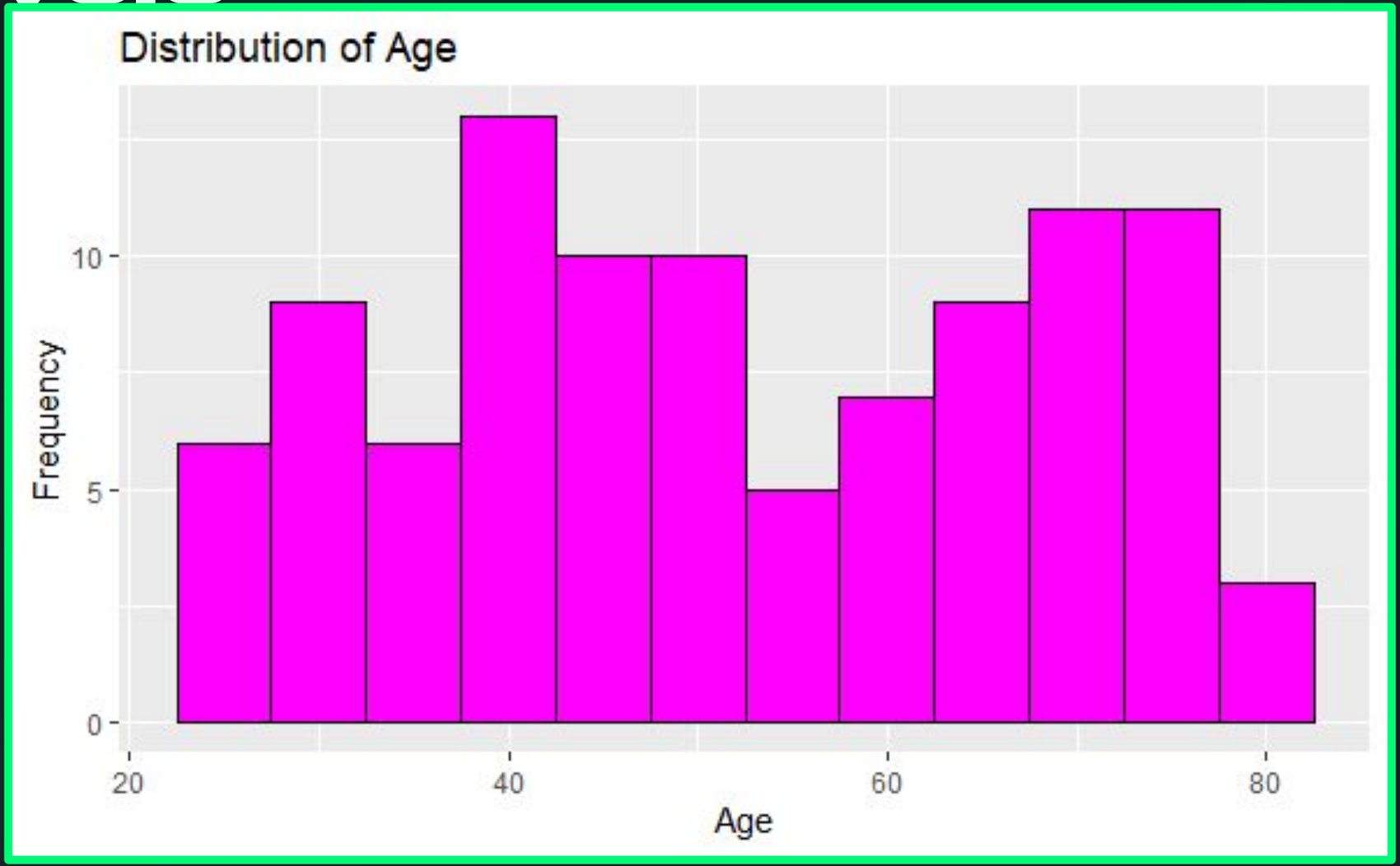
To check if there is any missing values in our	datasetsum(is.na(heart_data))

Libraries that I have used

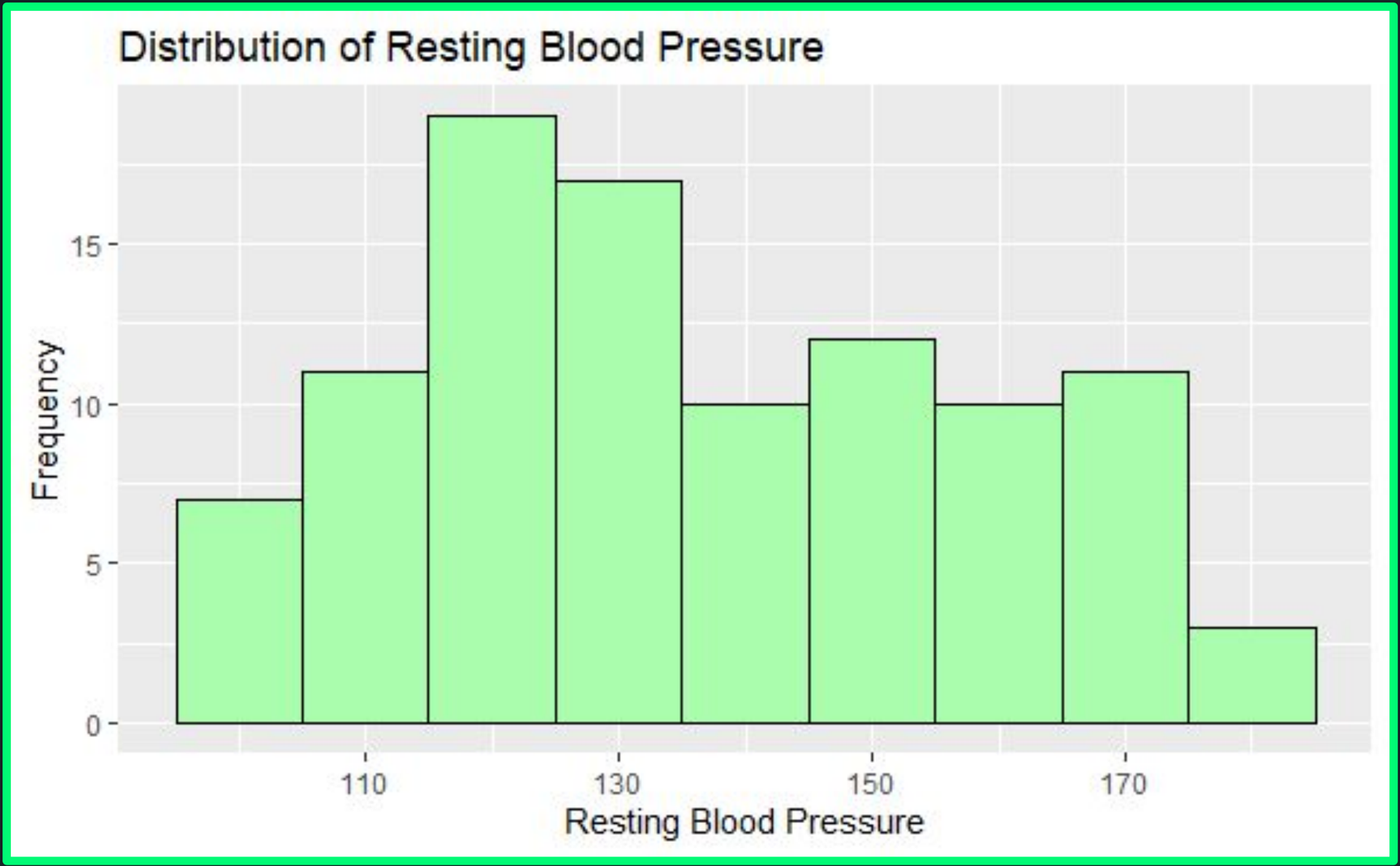
Tidyverse
ggplot2
Caret
Mass

Age Distribution Analysis

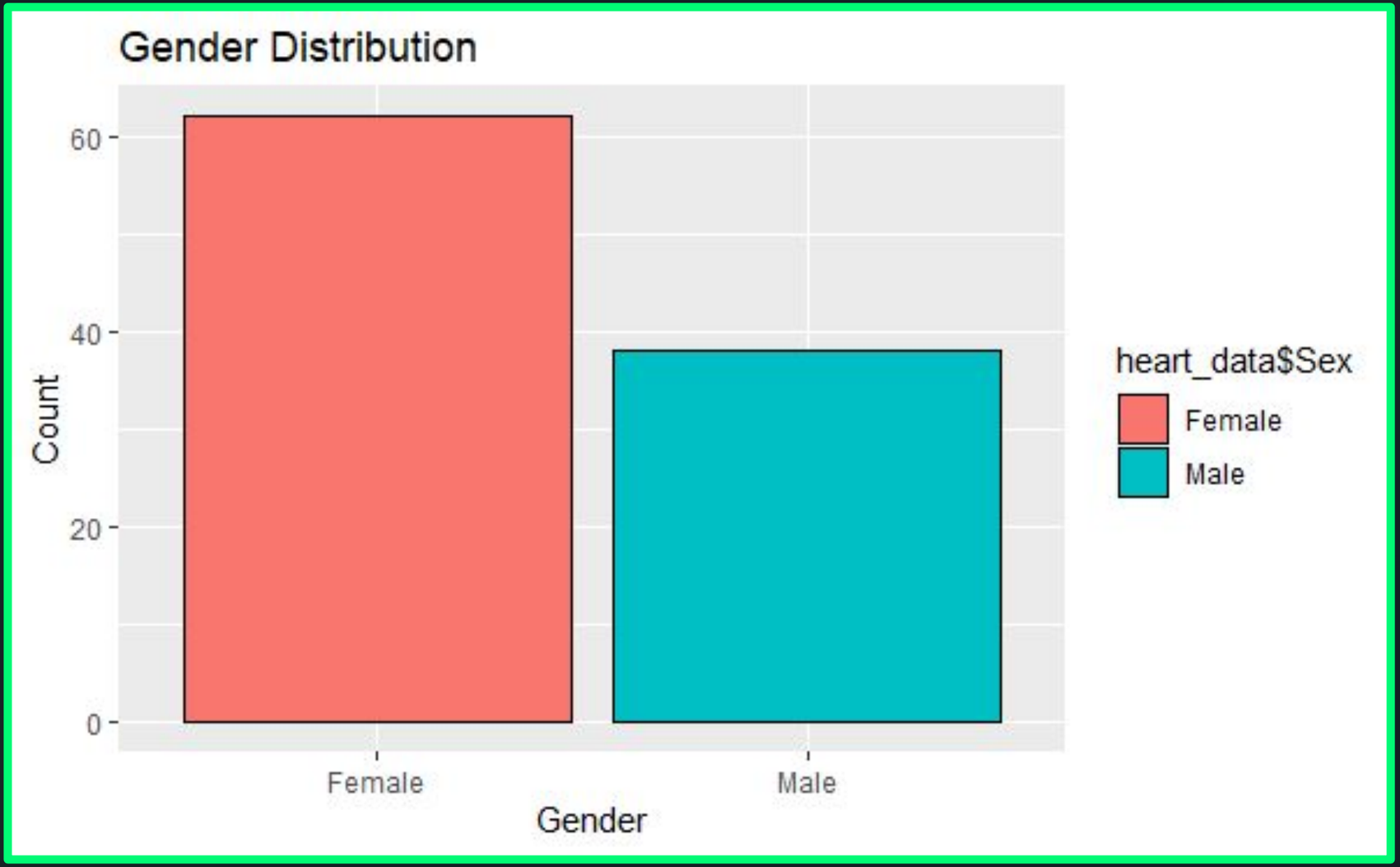
Understanding the age patterns in our heart attack risk dataset.



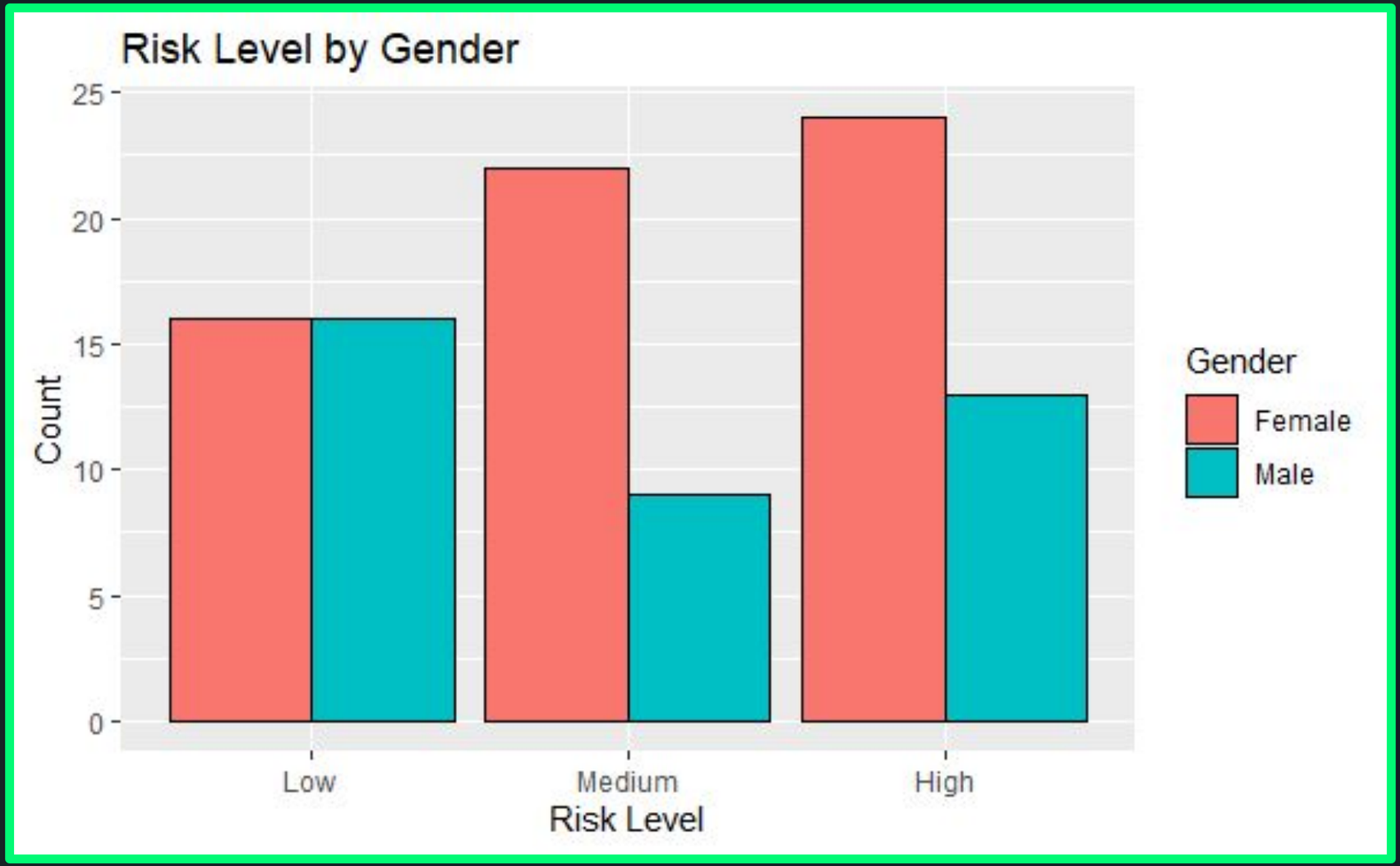
Distribution of Resting Blood Pressure



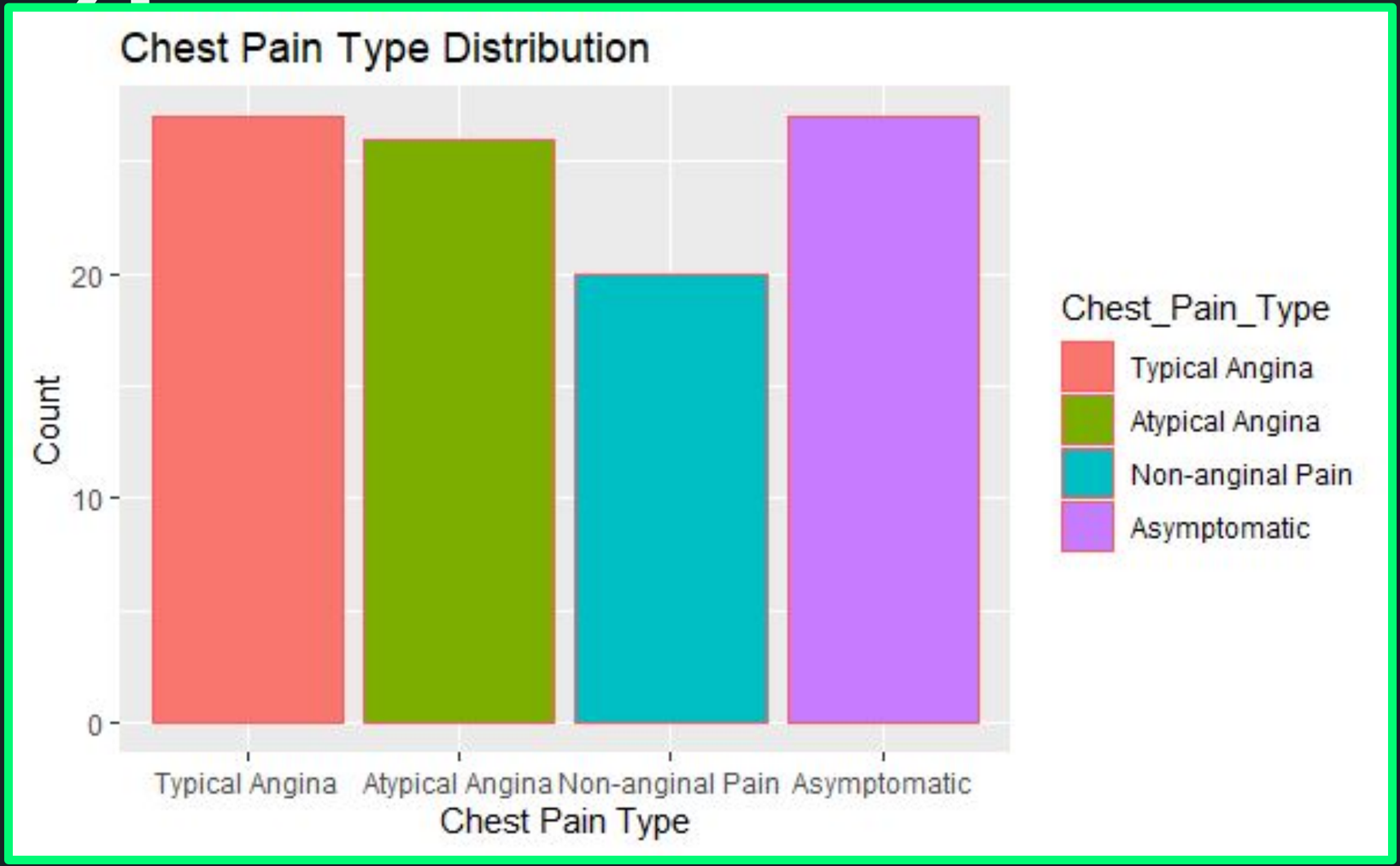
Gender Distribution in Dataset



Heart Attack Risk Level Distribution by Gender



Distribution of Chest Pain Types



Model Building

Variables Used

Our ordered logistic regression model incorporates 13 key clinical indicators, including vital signs, medical history, and test results. Each variable was carefully selected based on its clinical relevance to heart attack risk assessment.

- Patient demographics (Age, Sex)
- Cardiac indicators (Chest Pain, ECG results)
- Vital measurements (Blood Pressure, Heart Rate)
- Clinical tests (Cholesterol, Blood Sugar)

Model Implementation

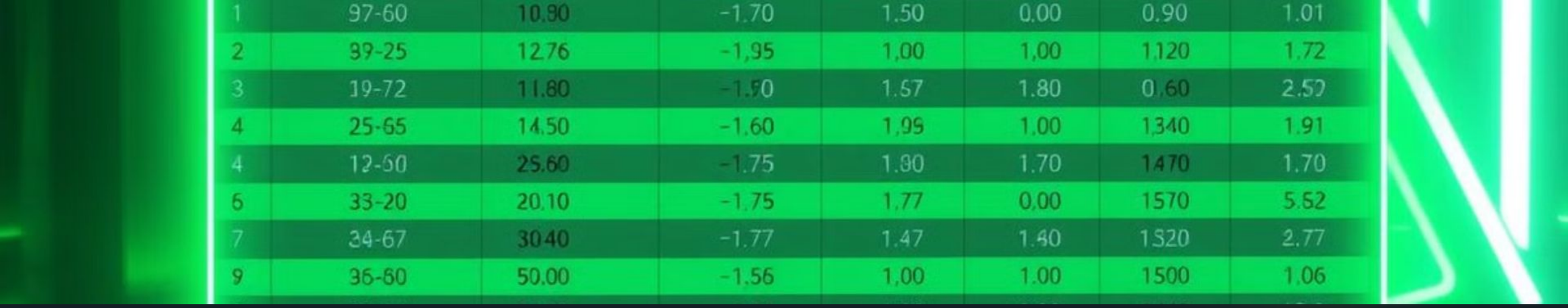
```
olr_model <- polr(
  Risk_Level ~ Age + Sex +
  Chest_Pain_Type + Resting_BP +
  Cholesterol + Fasting_Blood_Sugar +
  Resting_ECG + Max_Heart_Rate +
  Exercise_Induced_Angina +
  Oldpeak + Slope +
  Num_Vessels_Fluoroscopy +
  Thalassemia,
  data = train_data,
  Hess = TRUE)
```

Coefficient's understanding

- Odds Ratio (OR) tells us how the odds of being in a higher risk category change with each one-unit increase in a predictor.
- If **OR > 1**, it means the factor **increases** the chances of being in a higher risk category.
- If **OR < 1,** it means the factor **decreases** the chances of being in a higher risk category.
- If **OR = 1,** it means the factor has **no effect** on the odds of being in a higher risk category.

Model Coefficient

```
> #Step 8: Calculate Odds Ratios
> # Calculate and print odds ratios (exponentiate the coefficients)
> exp(coef(olr_model))
                          Age                          SexMale
                    1.0168935                        0.6604413
 Chest_Pain_TypeAtypical Angina Chest_Pain_TypeNon-anginal Pain
                    0.9041415                        0.6221899
    Chest_Pain_TypeAsymptomatic                       Resting_BP
                    0.8157588                        1.0027081
                  Cholesterol         Fasting_Blood_Sugar>120
                    1.0037997                        1.5244783
        Resting_ECGAbnormality          Resting_ECGHypertrophy
                    2.4506056                        1.5743851
              Max_Heart_Rate      Exercise_Induced_AnginaYes
                    1.0038973                        0.9538235
                     Oldpeak                        SlopeFlat
                    0.9626567                        0.9443154
            SlopeDownsloping         Num_Vessels_Fluoroscopy
                    1.8624856                        1.1596640
       ThalassemiaFixed Defect    ThalassemiaReversible Defect
                    0.3588716                        0.7911666
>
```



1	97-60	10.90	-1.70	1.50	0.00	0.90	1.01
2	39-25	12.76	-1,95	1,00	1,00	1,120	1.72
3	19-72	11.80	-1.50	1.57	1.80	0,60	2.50
4	25-65	14.50	-1.60	1,95	1.00	1,340	1.91
4	12-60	25.60	-1.75	1.90	1.70	1470	1.70
6	33-20	20.10	-1.75	1.77	0,00	1570	5.52
7	34-67	30.40	-1.77	1.47	1.40	1520	2.77
9	36-60	50.00	-1.56	1,00	1.00	1500	1.06

Results

- Some factors Increase the odds of being at higher risk (like abnormal ECG, High fasting blood sugar, and a downs loping slope).
- Others factors Increase the odds of being at higher risk (Like being male, having non aginal pain and having fixed Thalesemia defect)
- And many predictors variable does not show statistically significant effect (Like Age, Sex, Chest pain type)

Thank You